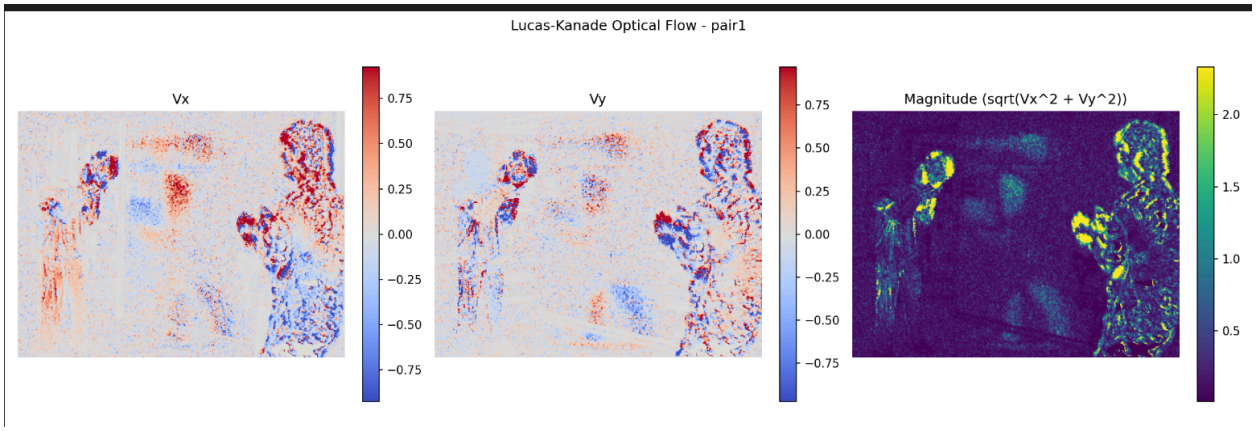
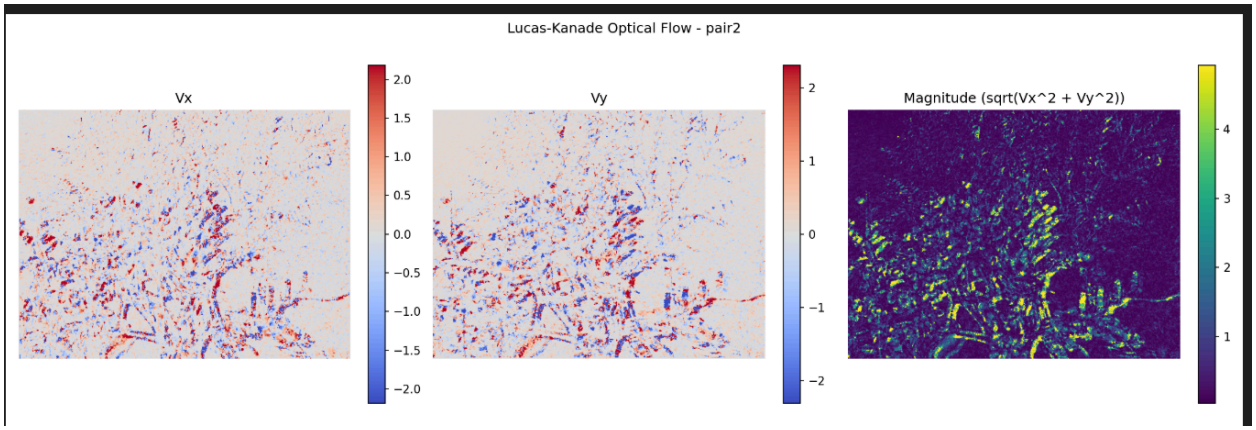
CS 410 Computer Vision and Deep Learning
Programming Assignment #2 Write-Up
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For this project, I've implemented the Lucas-Kanade optical flow method. The main purpose of this method is to estimate the motion vector at every pixel. First, I convolved the first frame with the Sobel-style kernels. The temporal gradient is the difference between the two frames/images.

The program was going all together; I have the two pairs of images that we compared and will talk about them below. I did notice that in the flat areas, like the blank wall, the pixels become singular, kinda combined. This gives unreliable results for the Vx and Vy values. I'm pretty sure this is a limitation of this algorithm. It can't tell and grab on some structure, so we just leave them as zero motion.



For this pair of images, it's two people playing basketball. The flow was focused on the ball and the moving hands. It didn't focus on the rest of the room. The magnitude of motion is shown in the light yellowish green dots. The room didn't have much magnitude, and actually close to zero magnitude. The focus is on the people and the ball, which highlights them more. This is good because we know it's the moving objects.



For this pair of images, there was some bush. The motion in these two frames is a bit more spread out. It is trying to capture the motion of the leaves. There is no certain region that the motion is focused on, but the whole image is trying to be detected.